Document Control

Title

Guidelines for managing hypercapnic respiratory failure with Non-Invasive Ventilation

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Superseded Documents

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- Glossop Ward Nursing Staff
- Emergency Department
- ITU
- Medical Consultants
- Respiratory Specialist Nurses

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- CRG

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1. Purpose

1.1. The purpose of this document is to provide clinical guidance to assist in the management of patients with hypercapnic respiratory failure who require ward based non-invasive ventilation (NIV)

1.2. The policy applies to in-patients with hypercapnic respiratory failure

1.3. Implementation of this policy will ensure that:

- Patients are safely managed and receive adequate treatment for their hypercapnic respiratory failure
- It is recommended that patients who require NIV are managed on a respiratory ward. In the event of ward closure options need to be discussed with the involvement of the infection control team

2. Definitions

2.1. AHRF - Acute Hypercapnic Respiratory Failure

2.2. NIV – Non Invasive Ventilation

2.3. EPAP – Expiratory Positive Airway Pressure

2.4. IPAP – Inspiratory Positive Airway Pressure

2.5. COPD – Chronic Obstructive Pulmonary Disease

2.6. OHS – Obesity Hypoventilation Syndrome

2.7. KS – Kyphoscoliosis

2.8. NMD – Neuromuscular disease

2.9. ABG – Arterial Blood Gas

3. Responsibilities

3.1. Medical staff

The Doctor or senior advanced practitioner commencing NIV is responsible for:

- Identifying a patient has AHRF
- Ensuring the indication for NIV is correct
- Ensuring there are no contraindications to NIV
- Contacting the respiratory Ward nursing staff to inform them of patient
- Giving nursing staff guidance on treatment targets
- Ensuring a treatment escalation plan is documented prior to commencing NIV
- Reviewing treatment efficacy and advising treatment changes
- Referring patient to respiratory team within 2 hours of admission or at start of next working day.

3.2. ITU referral

Consideration is required for patients who may be appropriate for ITU review. Patients with pH < 7.25, and PaCO2 > 6.0, escalation of care should be discussed with critical care at the time of or soon after commencement of NIV (NIV should not be delayed).

Patients who meet the critical care criteria for admission should be referred to the intensivist team

NIV may be considered as a ceiling of care for these patients, advice can be sought from the Respiratory Physicians (working hours 9am – 5pm) or from ITU.

3.3. Nursing Staff

The clinical environment for NIV should facilitate enhanced nursing and monitoring facilities

The Nurses managing the NIV are responsible for:

- Ensuring they are familiar with the guidelines for managing hypercapnic respiratory failure
- Setting up the equipment
- Monitoring the patient
- Ensuring the ward is adequately staffed
- Ensuring they have undertaken a trust competency to be able to manage the patients care

4. Documentation

4.1. NIV documentation to be completed in full can be found in appendix 1 and to follow the patient through their admission.

5. Indications for ward based NIV

5.1. COPD

pH <7.35 **and**

pCO2 >6.5

After initial 1 hour period of optimal medical management including controlled oxygen therapy **unless** initial pH <7.25 in which case NIV should be started immediately

5.2. Neuromuscular Disease

Respiratory illness **+/-** RR >20 with known/suspected respiratory muscle weakness **or**

pH <7.35 and pCO2 >6.5

5.3. Obesity

pH <7.35, pCO2 >6.5 and RR >23 **or**

pCO2 >6.0 and daytime somnolence

6. Ward NIV not indicated – consider ITU referral

6.1. Asthma

6.2. Pneumonia- Consider high flow nasal oxygen

6.3. Other causes of AHRF eg drug overdose, imminent death

Where NIV has been commenced in the Emergency Department and diagnosis is not COPD, MND or OHS the patient must be referred to HDU by the emergency department for ongoing care

See flow diagram in appendix 2 for NIV indication

7. Contraindications to NIV

7.1. Absolute

- Respiratory arrest/peri-arrest
- Severe facial deformity
- Facial burns
- Fixed upper airways obstruction

7.2. Relative – will need enhanced observation, consider HDU

- Undrained pneumothorax
- GCS <8 where airway is compromised
- pH <7.15 (or <7.25 with other adverse features)
- Confusion/agitation (that would affect ability to co-operate with treatment)
- Cognitive impairment
- Uncontrollable vomiting
- If NIV is the ceiling of care it may be used where adverse factors are present. The patient will need close observation for complications of treatment.

8. Setting up NIV

NIV should only be started in ED, HDU or on an appropriate respiratory unit Following an ABG. NIV should not be commenced from a venous blood gas

The respiratory ward must be fully staffed to accept a patient for NIV (5 trained nurses on dayshift, 4 trained nurses on nightshift). This enables the close monitoring of patients starting NIV.

If the respiratory Ward is understaffed patients requiring NIV may need to be admitted to HDU even if their ceiling of treatment is for ward based care. This is to enable safe nursing. A risk assessment should be undertaken by the medical team in consultation with ITU if this situation occurs.

When the respiratory Ward is closed patients on NIV may be managed on HDU or a secondary ward who can be supported by the respiratory ward staff.

An escalation plan should be documented **prior** to commencing NIV

An NIV checklist must be completed and placed in the patient's notes

NIV should be started within 1 hour of the ABG confirming it is required

8.1. Mask

A full face mask should be used as the first interface, sized appropriately to the patient

Masks for home NIV machines are not compatible with the in-patient machines and **must not** be used

Avoid over tightening of the mask. This can cause nasal ulceration which can lead to NIV failure

In relation to COVID 19 the machine must be turned off before the removal of any mask and the mask reapplied prior to switching the machine on

8.2. Other care

For COVID 19 the bacterial filter needs to be placed between the mask and the circuit (which is above the exhalation port)

Provide care to eyes and mouth.

Breaks from NIV should be made for drinks, drugs, physiotherapy, meals etc. Please refer to point 8.11 when mask is removed

Assess and monitor oral and dietary intake

All patients receiving NIV therapy should have a fluid balance chart to assist hydration assessment

NIV should be interrupted for the administration of nebulised bronchodilators

Administration of in line nebulisers via a standard nebuliser device is ineffective as the patient will only receive a small dose of the intended drug therapy

If the patient is unable to tolerate a break in NIV for nebuliser therapy continue NIV and seek senior advice.

Where an aerosol device is available (e.g. aeroneb in critical care) this may be used within the circuit as per manufacture guidelines.

8.3. Supplementary Oxygen

Supplemental oxygen can be added via port on the mask

If >4L oxygen is required use the V90 machine with piped oxygen

Oxygen to be titrated to prescribed oxygen saturation

A humidified NIV system is not routinely used

8.4. Pressure settings

Initial EPAP: 3cmH2O (unless OSA when may be higher)

Initial IPAP: 15 (20 if pH <7.25)

Up-titrate IPAP over 30 min to target IPAP 20-30 cmH2O

Aim for pressure sufficient to augment chest movement and slow respiratory rate

Higher EPAP may be needed in severe OHS and chest wall deformity

8.5. Back up rate

A back up rate of 8-12 should be used

9. Monitoring patients on NIV

9.1. Patients should be on continual saturation monitoring whilst on NIV

9.2. Target saturations should be prescribed as 88-92%

9.3. Continuous ECG monitoring must be used for the first 12 hours of NIV treatment and continued if there is ongoing tachycardia (pulse >120) or hypotension (systolic BP <100)

9.4. NIV settings and use must be documented on an NIV monitoring form

9.5. ABG should be repeated to monitor pH and pCO2 within 1 hour of achieving target settings then again at 4 hours and 12 hours

9.6. Patient should use NIV as much as tolerated in the first 24 hours

9.7. If a patient is not responding to NIV consider

9.7.1. Synchronisation
9.7.2. Mask fit/leak
9.7.3. Exhalation port patency
9.7.4. Adequate treatment of underlying condition (physiotherapy/bronchodilators)
9.7.5. HDU/ITU escalation

10. Weaning from NIV

10.1. Weaning from NIV can commence once arterial blood gases return to acceptable range set by medical team

10.2. Assess readiness for weaning on a daily basis

10.3. Aim for normalisation of pH and pCO2

10.4. Patients with chronic hypercapnia may not achieve a normal pCO2

10.5. Taper use of daytime NIV with breaks of 2-3 hours, gradually increasing duration of breaks

10.6. Discontinue nocturnal NIV when self-ventilating pCO2 is within agreed target range

11. Decontamination and cleaning

The machine should be wiped over/ cleaned daily in accordance with Trust cleaning standards. Following use the machine should be cleaned and decontaminated following infection control and manufacturer's instructions.

There is a reusable foam inlet filter, required to protect the ventilator from dirt and dust. This requires cleaning every week and replacing periodically (recommended every 6 months) and/or if damaged.

In relation to COVID 19 no additional cleaning and filter changes needs to be made from the normal process

12. Monitoring Compliance with and the Effectiveness of the Guideline

12.1. Education

- All Staff in ED and the respiratory ward who have contact with NIV patients must receive NIV training. All nursing staff to undertake a competency assessment

12.2. Process for Implementation and Monitoring Compliance and Effectiveness

This policy will be monitored and updated to ensure that best practice and new evidence will be considered

Details of monitoring

- Audit of compliance of NIV proforma completion
- Annual audit undertaken on the patients who require NIV with a patient register being completed
- Audit on the completion of the daily NIV observation chart

13. References (Optional) Staff

- BTS/ICS Guidelines for the Ventilatory Management of Acute Hypercapnic Respiratory Failure in Adults
- Inspiring Change – NCEPOD report into use of acute non-invasive ventilation

14. Associated Documentation (Optional)

- NIV checklist/ flow diagram
- NIV monitoring form
- NIV competency assessment

Indications for NIV

COPD
pH <7.35
pCO2 >6.5
RR>23
If persisting after bronchodilators and controlled oxygen therapy

Neuromuscular disease
Respiratory illness with RR > 20 if usual VC <1L even if pCO2< 6.5
Or
pH < 7.35 and pCO2> 6.5

Obesity
pH <7.35, pCO2>6.5, RR>23
Or
Daytime pCO2> 6.0 and somnolent

Contraindications for NIV

Absolute
Severe facial deformity
Facial burns
Fixed upper airway obstruction

Relative
pH<7.15
(pH<7.25 and additional adverse feature)
GCS <8
Confusion/agitation
Cognitive impairment (warrants enhanced observation)

Indications for referral to ICU
AHRF with impending respiratory arrest

NIV failing to augment chest wall movement or reduce pCO2

Inability to maintain Sao2 > 85-88% on NIV

Need for IV sedation or adverse features indicating need for closer monitoring and/or possible difficult intubation as in OHS, DMD.

NIV Not indicated
Asthma/Pneumonia
Refer to ICU for consideration IMV if increasing respiratory rate/distress
or
pH <7.35 and pCO2 >6.5

NIV SETUP

Mask
Full face mask (or own if home user of NIV)

Initial Pressure settings
EPAP: 3 (or higher if OSA known/expected)

IPAP in COPD/OHS/KS 15 (20 if pH <7.25)

Up titrate IPAP over 10-30 mins to IPAP 20—30 to achieve adequate augmentation of chest/abdo movement and slow RR

IPAP should not exceed 30 or EPAP 8* without expert review

IPAP in NM 10 (or 5 above usual setting)

Backup rate
Backup Rate of 16-20. Set appropriate inspiratory time

I:E ratio
COPD 1:2 to 1.3
OHS, NM & CWD 1:1

Inspiratory time
0.8-1.2s COPD
1.2-1.5s OHS, NM & CWD

Use NIV for as much time as possible in 1st 24hours.
Taper depending on tolerance & ABGs over next 48-72 hours

SEEK AND TREAT REVERSIBLE CAUSES OF AHRF

* Possible need for EPAP > 8
Severe OHS (BMI >35), lung recruitment eg hypoxia in severe kyphoscolios, oppose intrinsic PEEP in severe airflow obstruction or to maintain adequate PS when high EPAP required

NIV Monitoring

Oxygenation
Aim 88-92% in all patients

Note: Home style ventilators CANNOT provide > 50% inspired oxygen.

If high oxygen need or rapid desaturation on disconnection from NIV consider IMV.

Red flags
pH <7.25 on optimal NIV
RR persisting > 25
New onset confusion or patient distress

Actions
Check synchronisation, mask fit, exhalation port : give physiotherapy/bronchodilators, consider anxiolytic

CONSIDER IMV

NIV weaning
Aim to normalise PH and co2

Taper daytime use with breaks of 2-3 hours and gradually increase

Discontinue nocturnal NIV when self-ventilating pCONon Invasive Ventilation Proforma

Initial NIV Prescription and Change to Settings

	Nursing competencies	Assessment criteria	Nurses signature	Assessors signature	Date achieved
1	Demonstrate an understanding of the anatomy and physiology of the respiratory system	Identify the main structures of the respiratory system Discuss the position of these structures in relation to their function Discuss the mechanism of normal breathing by negative pressure Identify the difference between type 1 and type 2 respiratory failure and discuss why some patients may be oxygen sensitive and target saturations may need to be altered			
2	Identify the contraindications for NIV (BIPAP)	Be able to identify the contraindications for commencing NIV			
3	Correctly assemble and prepare the equipment	Ensure bipap machine is correctly plugged in and switched on Ensure filters are changed as required Identify and measure the patient for the correct mask Identify exhalation port on hose Check mask for leaks Check patients breathing with the machine			
4	Explain what is needed prior to starting NIV therapy, check capacity, consent and check for medical escalation plan	Check NIV prescription and proforma has been completed Any issues with consent/capacity? Ensure chest xray has been requested and completed to rule out pneumothorax (DO NOT DELAY TREATMENT WAITING FOR CXR) Ensure BIPAP is being used as per local policy			
5	Provide an explanation to the patient and to the next of kin	Provide advice to the patient on what will happen, the type of mask to be used and the effects of treatment			
6	Explain and implement the safety measure required for BiPAP	Monitoring of patient with cardiac monitor with continual saturation monitoring Oxygen therapy prescribed Understand the alarms on the machine			
7	Demonstrate the ability to commence the treatment (on a patient) and plan within the first hour, with underpinning rationale for actions	Understanding of acute hypercapnia and type 2 respiratory failure Understanding of IPAP setting Understanding of EPAP settings			

		Understand how to increase the setting for IPAP and EPAP parameters Demonstrate how to monitor a patients respiratory rate and o2 saturations Record accurately how to record an EWS			
8	Accurately record the treatment of the patient, changes in the patient's condition and maintains appropriate documentation throughout the patient' treatment	Able to demonstrate how to complete the EWS and when to escalate appropriately Demonstrate accurate documentation on the NIV daily chart Demonstrate clear documentation as per NMC guidelines. Demonstrate the ability to be able to clearly and accurately document an ABG			
9	Ensure medication is administered to the patient receiving BiPAP therapy and discuss actions to take if the patient desaturates on mask removal.	Demonstrate the understanding of using other therapies eg nebuliser (In line and air driven), steroids, use of antibiotics,			
10	Hydration & fluid balance	Show an understanding of how to ensure patients receive adequate hydration and nutrition while receiving bipap Explain how to facilitate breaks for fluid, nutrition and medication Identify how to record nutritional and fluid balance on these patients			
11	Complications with NIV	Discuss how you would manage non-compliance with treatment How would you manage patients coming to the end of life			
12	Maintaining of machines	What filters do you need to change? How often do they require changing Cleaning the machine			